

Math 4317 : Real Analysis I

Mid-Term Exam 1

25 September 2012

Instructions: Answer all of the problems.

Definitions (2 points each)

1. State the definition of a metric space.

A metric space is a space with distance defined within a point set, X . This distance follows three axioms:

$$d(x, y) > 0$$

$$d(x, y) = 0 \iff x = y$$

$$d(x, z) \leq d(x, y) + d(y, z)$$

2. State the definition of an open set in a metric space.

An open set in a metric space occurs when an open ball $B(x, r) \subseteq E \forall r > 0$.

3. State the definition of the greatest lower bound of a set of real numbers.

The greatest lower bound for a set X is when $\forall x \in X$, there exists a value $\alpha \leq x$. With α' being some other lower bound, we have $\alpha' < \alpha \leq x$.

4. State the definition for a set to be countable.

A set X is countable if there exists a bijection $f : X \rightarrow \mathbb{N}$

5. State the Least Upper Bound Property of $\mathbb{R}$.

True or False (1 point each)

1. If F is a set in a metric space that is not closed, then it is open.
2. If F is closed and K is compact, then $F \cap K$ is compact.
3. For any collection $\{G_\alpha\}$ of open sets, $\bigcup_\alpha G_\alpha$ is open.
4. For any finite collection $\{F_1, \dots, F_n\}$ of closed sets, $\bigcup_{j=1}^n F_j$ is closed.

5. Let (X, d) be a metric space. For $x \in X$ and $r > 0$ the set

$$B_r(x) = \{y \in X : d(x, y) < r\}$$

is open.

6. If $x, y \in \mathbb{R}^n$, then

$$\left| \sum_{j=1}^n x_j y_j \right| \leq \left(\sum_{j=1}^n x_j^2 \right)^{\frac{1}{2}} \left(\sum_{j=1}^n y_j^2 \right)^{\frac{1}{2}}.$$

7. In a metric space X , the set X is the only set that is both open and closed.
8. For any collection of compact sets $\{K_\alpha\}$, $\bigcup_\alpha K_\alpha$ is compact.
9. If E is a nonempty subset of $\mathbb{R}$ that is bounded from below, then $\inf E \in E$.

10. The set of all irrational numbers is countable.
11. A set is open if and only if it is equal to its interior.
12. A set is closed if and only if it contains its boundary.
13. A set in an arbitrary metric space is compact if and only if it is closed and bounded.
14. In a metric space (X, d) , one has that $\bigcap_{r>0} B_r(x) = \emptyset$.
15. Every infinite subset of a countable set is countable.

Proofs (15 pts each)

1. A set S in a metric space (X, d) is called *totally bounded* if for any $\varepsilon > 0$ there exists a finite subset T of S such that

$$S \subset \bigcup_{x \in T} B_\varepsilon(x).$$

Let $K \subset X$ be a compact subset of the metric space (X, d) . Prove that K is totally bounded.

2. Let A be a non-empty set of real numbers which is bounded below. Let

$$-A = \{-x : x \in A\}.$$

Prove that

$$\inf A = -\sup(-A).$$

3. Let $a \in \mathbb{R}$ with $a > -1$. Prove that $(1 + a)^n \geq 1 + na$ for all $n \in \mathbb{N}$.

4. Let

$$X = \left\{ x = (x_1, x_2, x_3, \dots) : x_i \in \mathbb{R} \text{ and } \sup_i |x_i| < \infty \right\}.$$

Show that (X, d) is a metric space when

$$d(x, y) := \sup_i |x_i - y_i|.$$

5. Show directly from the definitions that

$$G = \{(x, y) \in \mathbb{R}^2 : 4x^2 + y^2 < 1\}$$

is open in $\mathbb{R}^2$ with the metric $d((x, y), (u, v)) = \sqrt{(x - u)^2 + (y - v)^2}$.